IISE LSC Student Case Competition

Team Contact Information

University Name: Binghamton University

Department: Industrial and Systems Engineering (SSIE)

Competition Category: Graduate

Team Name: Team BingLogistics

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MEMORANDUM

To: IISE LSC Division Awards Committee
From: Team BingLogistics, Binghamton University
Date: February 20th, 2026

Subject: Annual Freight Transportation Estimate - Growing Pains Case Study

1. Problem Overview

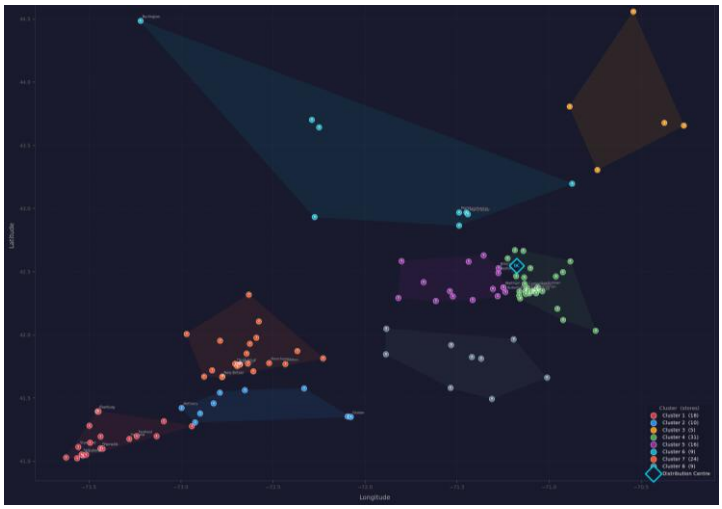
Northeastern Home Goods (NHG) operates 123 stores across six Northeastern states and currently uses a private fleet to deliver goods on a fixed weekly schedule during store hours (8:00 a.m. – 6:00 p.m.). Each store receives deliveries on specific weekdays that remain consistent every week. While this system provides stability, managing the growing fleet has become increasingly complex. NHG is considering outsourcing transportation to Massachusetts Area Distribution (MAD), which operates a single distribution center in Wilmington, Massachusetts. Before reviewing MAD's proposal, NHG requested an internal estimate of total annual freight miles assuming all stores are served from Wilmington DC, the fixed weekly schedule is maintained, deliveries occur only during operating hours, and orders from different weekdays cannot be consolidated onto the same route. The dataset includes average weekly freight volumes totaling 65,429 cubic feet across 262 deliveries and a complete distance matrix between the DC and stores. Routes must follow operational and regulatory constraints, including a 3,200 cubic feet trailer capacity, U.S. DOT limits of 11 driving hours or 14 on-duty hours with a required 10-hour break, a minimum 30-minute unloading time per stop, an average speed of 40 mph, and departure timing that ensures arrival at the first store exactly at 8:00 a.m. The objective is to design feasible daily routes that satisfy all constraints while minimizing total miles, providing an annual mileage estimate to evaluate MAD's proposal.

2. Methodology

2.1 Data Preparation

During the initial analysis, we observed that some stores had multiple delivery records on the same day. Since deliveries to the same store on the same day can be combined into a single optimized stop, we aggregated the total cubic feet required per store per day before routing. This ensured routing decisions were based on total daily demand at each location. Stores whose daily demand exceeded the 3,200 cubic feet trailer capacity, those deliveries were split into multiple visits within capacity limits. We also noted that on Tuesday, a DC-to-DC delivery record appeared and it was not considered for the routing schedule. Unloading time per stop was calculated as the greater of 30 minutes or 0.030 multiplied by the delivery volume. For each day, store demand data was cleaned, verified against the master distance matrix, and grouped by ZIP code. Travel times were calculated from the distance matrix using the given average speed. Geographic analysis of store locations relative to the Wilmington distribution center showed regional clusters as well as longer-distance deliveries. This provided insight into expected route lengths, truck utilization, and possible extended return trips.

Days of Week	Total Volume Freight	Total Deliveries	Unique Stores Delivery
Monday	10223	43	34
Tuesday	11537	58	40 (excluding DC delivery)
Wednesday	15192	50	39
Thursday	15009	63	45
Friday	13468	47	41



Clusters	No ‘of Stores
1	18
2	10
3	5
4	31
5	16
6	9
7	24
8	9

Fig 1: Store Location Clustering Summary: 8 Clusters Detected

2.2 Modeling Approach

The routing problem was modeled as a Capacitated Vehicle Routing Problem with Time Windows (CVRPTW). This formulation allows vehicle capacity limits, delivery time windows, and route duration constraints to be handled within a single structured optimization framework. CVRPTW is widely recognized in transportation research as the standard approach for realistic delivery routing problems. The baseline solution approach used Path Cheapest Arc (PCA) to construct an initial feasible routing plan. PCA is a greedy construction heuristic that builds routes step-by-step by repeatedly selecting the lowest-cost next connection between locations. To improve this initial solution, we applied Guided Local Search (GLS) as a metaheuristic improvement phase. GLS enhances local search by penalizing frequently used high-cost route features, allowing the algorithm to escape local optima and progressively reduce total travel distance while maintaining feasibility. This construct-then-improve strategy is well-supported in routing literature as a practical and scalable method for large operational datasets [1][2]. To enhance geographic efficiency, we performed hierarchical clustering on store coordinates prior to optimization. Hierarchical clustering groups nearby stores into natural service regions based on spatial proximity. The resulting clusters (Figure 1) were incorporated into the PCA–GLS–SoftClust model as a soft penalty in the objective function. Instead of enforcing strict geographic boundaries, the model discourages unnecessary cross-region travel while preserving flexibility, leading to more compact and operationally intuitive routes. In addition, we evaluated a second construction strategy, Parallel Cheapest Insertion (PCI). Unlike PCA, which extends routes sequentially, PCI builds multiple routes simultaneously by inserting customers into the most cost-effective position across routes. The PCI solution was then improved using the same GLS metaheuristic. Insertion-based methods are frequently reported in the literature as producing strong initial solutions, making them suitable for comparison against arc-based construction [3]. These models were selected because they are well-established in vehicle routing research, computationally efficient, and capable of producing high-quality, feasible solutions for large-scale daily delivery planning. The table below summarizes each model and its corresponding execution time

Model	Description	Execution Time
PCA-GLS (Baseline)	Path Cheapest Arc heuristic combined with Guided Local Search for route improvement	120.5 seconds
PCA-GLS-SoftClust	Path Cheapest Arc with Guided Local Search, incorporating a soft clustering mismatch penalty within the arc cost function	123.5 seconds
PCI-GLS	Parallel Cheapest Insertion heuristic integrated with Guided Local Search for enhanced route optimization	180.6 seconds

2.3 Assumptions

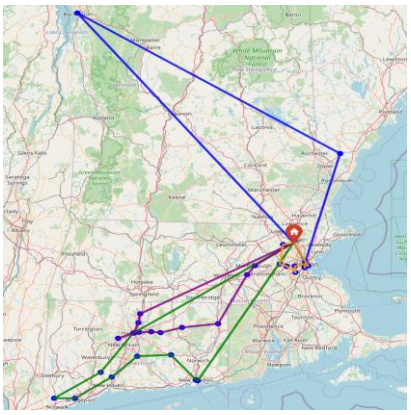
We built independent daily schedules with no cross-day consolidation, assuming a single depot (ZIP 01887) where all routes start and end. Travel times were derived from the ZIP distance matrix using a constant speed of 40 mph (1 mile = 1.5 minutes), with service allowed between 8:00 a.m. and 6:00 p.m., dispatch timed to reach the first stop at 8:00 a.m., and no loading time considered at the DC. Each truck has a hard capacity of 3,200 cubic feet, with daily store demand aggregated and split if it exceeds capacity. Only store unloading time was modeled, and simplified Hours-of-Service limits were enforced (11 driving hours, 14 on-duty hours), with routes split or 10 hour rest inserted if limits were exceeded; all trucks return to the DC, and location coordinates were assumed accurate for mapping and clustering.

3. Results

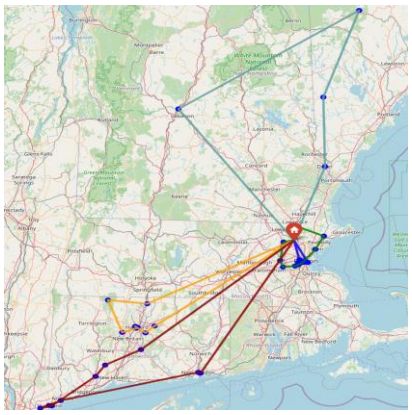
The table below summarizes the performance results of all the tested models

	PCA-GLS		PCA-GLS-SoftClust		PCI-GLS	
	Req. Trucks	Total Miles	Req. Trucks	Total Miles	Req. Trucks	Total Miles
Monday	4	1301	5	1331	4	1301
Tuesday	5	1345	5	1355	5	1345
Wednesday	6	1160	6	1165	5	1373
Thursday	6	1392	6	1412	5	1455
Friday	5	1160	6	1195	5	1160

	PCA-GLS	PCA-GLS-SoftClust	PCI-GLS
Total Unique Trucks Required per Week	8	8	7
Average Days Occupied per Truck	3	3	3
Average Truck Utilization	74%	70%	83%



Monday



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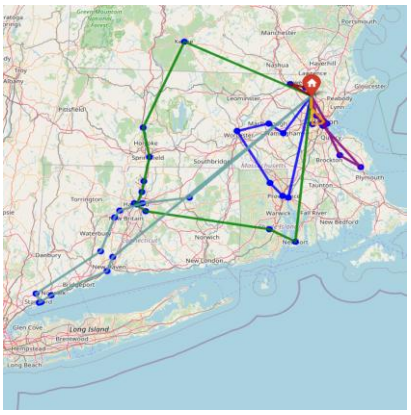
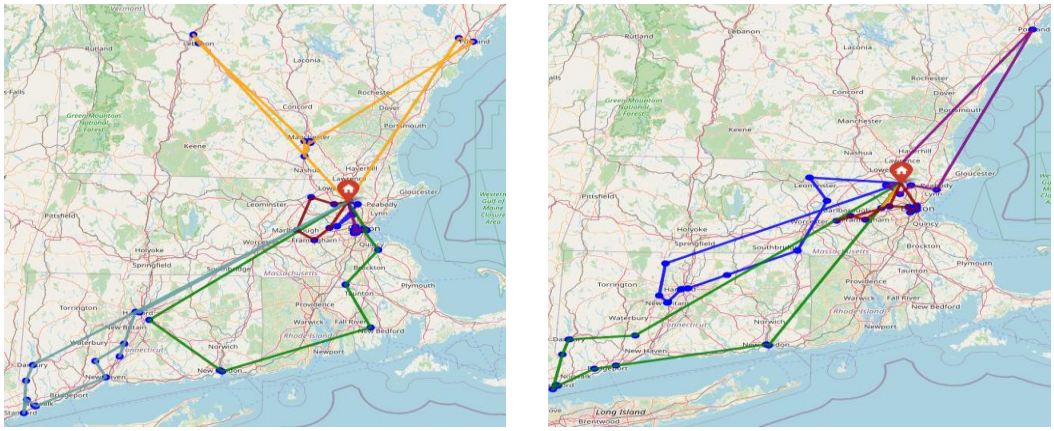


Wednesday



Thursday Friday

Fig 2: Weekly Truck Routing Results for the PCA-GLS Model

4. Recommendation

Based on our weekly routing analysis, we recommend using the **PCA–GLS optimization model** as the official baseline for estimating annual freight miles and evaluating MAD's outsourcing proposal. This approach consistently produced feasible, regulation-compliant schedules while minimizing total mileage under capacity, time-window, and Hours-of-Service constraints. Across all weekdays, PCA–GLS delivered the lowest mileage results while maintaining stable truck requirements. In comparison, the PCA–GLS–SoftClust model improved geographic compactness and operational clarity but slightly increased total miles and, on some days, required an additional truck. Therefore, clustering should be considered an operational enhancement rather than the cost baseline. The PCI–GLS variant produced mixed results. While it reduced truck requirements on certain days, this reduction came at the expense of increased total mileage. This indicates a trade-off between minimizing fleet size and minimizing total travel distance. Since the primary objective defined in the case is minimizing annual freight miles, PCA–GLS remains the most appropriate baseline model. The optimized PCA–GLS results show a total weekly mileage of **6,358 miles**, which translates to approximately **330,616 miles annually**. The maximum fleet requirement is **six** trucks per day and **eight** unique trucks per week, including a mix of day cabs and three sleeper cabs to support longer routes. The average truck utilization is approximately **74%**, indicating relatively strong asset usage with some opportunity for marginal improvement. We recommend using this internally derived annual mileage estimate as the quantitative benchmark when assessing the fairness and competitiveness of MAD's outsourcing proposal, while recognizing that alternative models provide insight into trade-offs between fleet size, route compactness, and mileage efficiency.

5. References

[1] Beullens, P., Muyldermans, L., Cattrysse, D., & Van Oudheusden, D. (2003). A guided local search heuristic for the capacitated arc routing problem. *European Journal of Operational Research*, *147*(3), 629–643. doi:10.1016/S0377-2217(02)00334-X

[2] Voudouris, C., Alsheddy, A., & Alhindi, A. (2025). Guided Local Search. In R. Martí, P. M. Pardalos, & M. G. C. Resende (Eds), *Handbook of Heuristics* (pp. 427–467). doi:10.1007/978-3-032-00385-0_2

[3] Chapleau, L., Ferland, J. A., Lapalme, G., & Rousseau, J.-M. (1984). A parallel insert method for the capacitated arc routing problem. *Operations Research Letters*, *3*(2), 95–99. doi:10.1016/0167-6377(84)90049-X